

SIMATS ENGINEERING

ASSESSMENT TOOL I — Uncertainty Visualization Lab

Course: Data Handling and Visualization	Code: DSA06	CO: C05
Duration: 180 min	Total Marks: 100	Reg No : 192324298 Name: Akash S

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Understanding of Uncertainty 25%	Visualization of Uncertainty 25%	Application of Visualization Principles 20%	Organization 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Bootstrapping and Confidence Intervals – Point Estimate

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data: 31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: resampling, point estimate, confidence interval, histogram/density plot, interpretation.

Answer Q1:

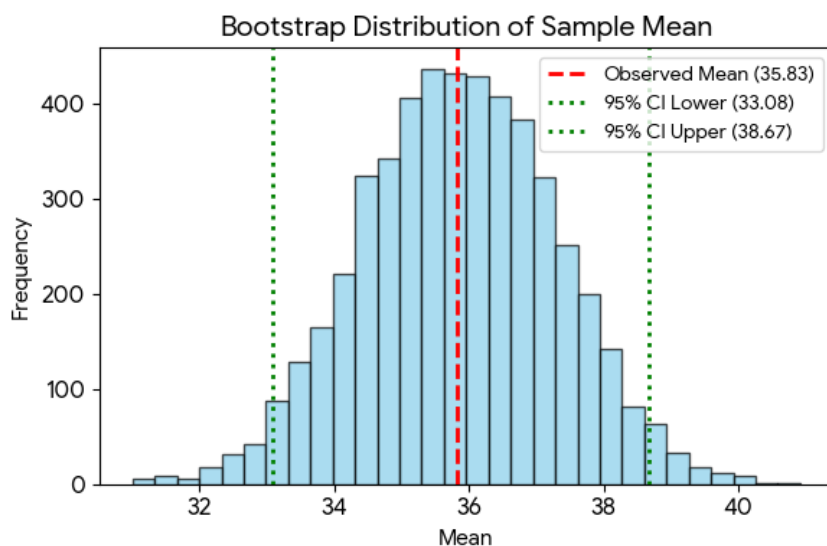
```
# R Code
library(boot)
library(ggplot2)

data <- c(31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36)
sample_mean <- mean(data)

# Bootstrap function
mean_func <- function(d, i) { return(mean(d[i])) }
set.seed(42)
boot_out <- boot(data, mean_func, R = 5000)

# Confidence interval
boot_ci <- boot.ci(boot_out, type = "perc")
print(boot_ci)

# Visualization
boot_data <- data.frame(means = boot_out$t)
ggplot(boot_data, aes(x = means)) +
  geom_histogram(binwidth = 1, fill = "skyblue", color = "black", alpha = 0.7) +
  geom_vline(aes(xintercept = sample_mean), color = "red", linetype = "dashed", size = 1) +
  geom_vline(aes(xintercept = boot_ci$percent[4]), color = "green", linetype = "dotted", size = 1) +
  geom_vline(aes(xintercept = boot_ci$percent[5]), color = "green", linetype = "dotted", size = 1) +
  labs(title = "Bootstrap Distribution of Sample Mean", x = "Mean", y = "Frequency")
```



Interpretation: The histogram shows the bootstrap sampling distribution of the mean. The red dashed line represents the observed sample mean, and the green dotted lines represent the 95% confidence bounds. This interval indicates where we expect the true population mean to lie with 95% confidence based on our sample.

Q2. Uncertainty in Trend Estimation

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band. Explain how the uncertainty changes across the range of advertising expenditure.

Data: Advertising = 10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40;

Sales = 48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

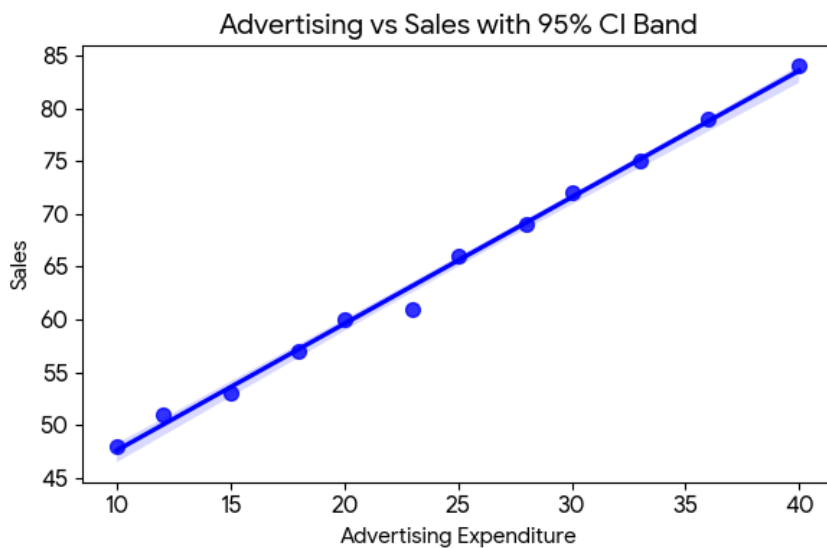
Expected R tasks: `lm()`, prediction interval/confidence interval, `ggplot2` visualization, trend interpretation.

Answer Q2:

```
# R Code
library(ggplot2)
adv <- c(10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40)
sales <- c(48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84)
df <- data.frame(adv, sales)

# Linear model
model <- lm(sales ~ adv, data = df)
summary(model)

# Plot
ggplot(df, aes(x = adv, y = sales)) +
  geom_point(size = 3, color = "blue") +
  geom_smooth(method = "lm", level = 0.95, color = "darkblue", fill = "lightblue") +
  labs(title = "Advertising vs Sales with 95% CI Band", x = "Advertising Expenditure", y = "Sales")
```



Interpretation: The light blue band indicates the 95% confidence interval for the regression line. The uncertainty is narrowest near the mean of the advertising expenditure and widens at the extremes. This implies predictions made near the edges of the data range are less certain than those made near the center.

Q3. Overplotting, Jittering and Transparency

A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation.

Data: Hours = 2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,7,8,8,8,8,9,9,9,10,10;

Scores = 45,52,52,58,60,60,64,66,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94

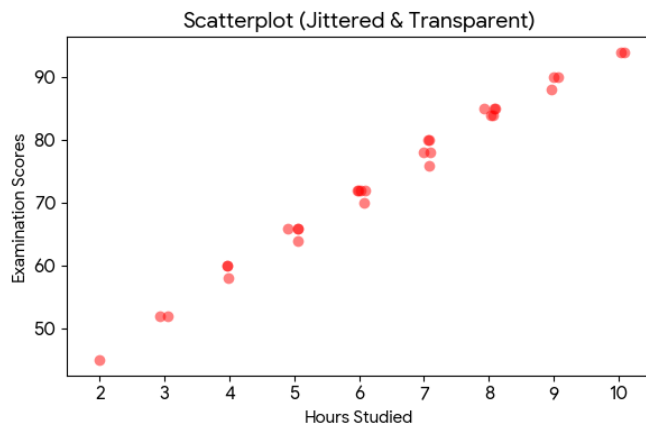
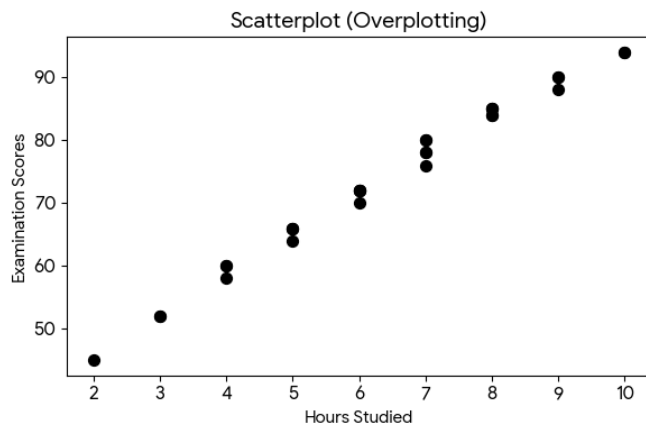
Expected R tasks: `geom_point()`, `alpha`, `position_jitter()`, comparison of overplotting.

Answer Q3:

```
# R Code
library(ggplot2)
hours <- c(2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,7,8,8,8,8,9,9,9,10,10)
scores <- c(45,52,52,58,60,60,64,66,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94)
df3 <- data.frame(hours, scores)

# Plot 1: Overplotted
p1 <- ggplot(df3, aes(x = hours, y = scores)) + geom_point(size = 3) + labs(title = "Without Modification")
print(p1)

# Plot 2: Jitter and Transparency
p2 <- ggplot(df3, aes(x = hours, y = scores)) +
  geom_jitter(width = 0.2, height = 0.5, alpha = 0.5, size = 3, color = "red") +
  labs(title = "With Jitter and Transparency")
print(p2)
```



Comparison: In the first plot, many data points are hidden because they share identical x and y values (overplotting), which masks the true density and cluster size. In the redesigned plot, jittering adds small random noise to the positions, and transparency allows overlapping points to appear darker. This reveals the true distribution and concentration of students' hours and scores.

Q4. 2D Histogram and Density Estimation

A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

Data: Height = 150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186;

Weight = 48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88

Expected R tasks: 2D binning, density estimation, contour/density layers, interpretation of concentration.

Answer Q4:

R Code

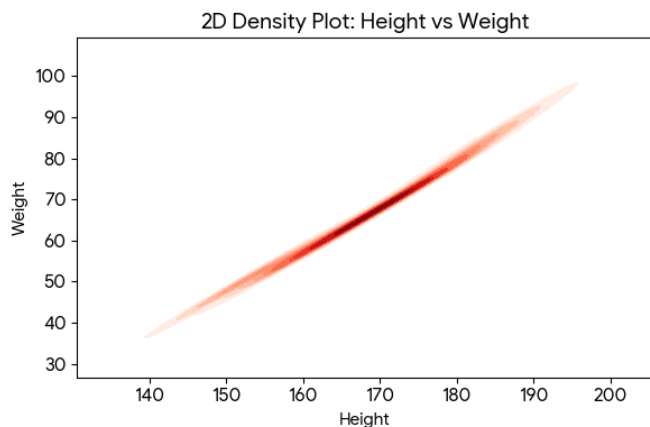
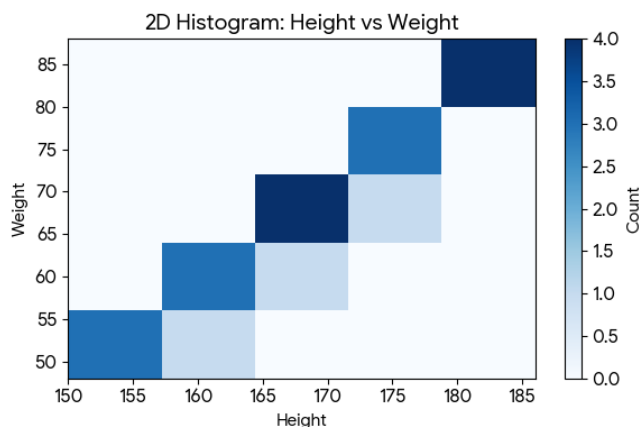
```
library(ggplot2)
height <- c(150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186)
weight <- c(48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88)
df4 <- data.frame(height, weight)
```

2D Histogram

```
ggplot(df4, aes(x = height, y = weight)) +
  geom_bin2d(bins = 5) +
  scale_fill_continuous(type = "viridis") +
  labs(title = "2D Histogram")
```

2D Density

```
ggplot(df4, aes(x = height, y = weight)) +
  geom_density_2d_filled(alpha = 0.8) +
  labs(title = "2D Density Visualization")
```



Justification: The 2D density plot communicates concentration more clearly because it smooths the discrete data into continuous contour levels, making the central mass and gradients visually distinct. The 2D histogram's blocky bins are heavily influenced by the arbitrary choice of bin boundaries, sometimes breaking up clusters unnaturally, particularly for small datasets.

Q5. Color Scales, Accessibility and Misleading Visualization

The following category-wise satisfaction scores must be visualized in R. Create two plots: (1) a plot using a nonmonotonic/diverging color scheme that could mislead interpretation, and (2) a redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale. Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data: Category = A,B,C,D,E,F,G;

Satisfaction = 42,48,55,61,67,74,82

Expected R tasks: color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.

Answer Q5:

R Code

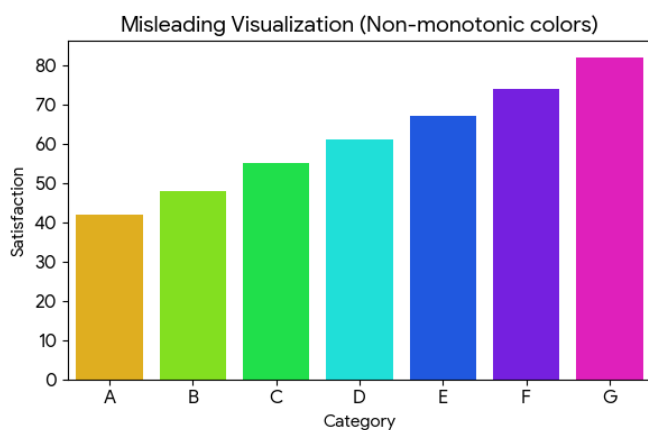
```
library(ggplot2)
cat <- c('A','B','C','D','E','F','G')
sat <- c(42,48,55,61,67,74,82)
df5 <- data.frame(cat, sat)
```

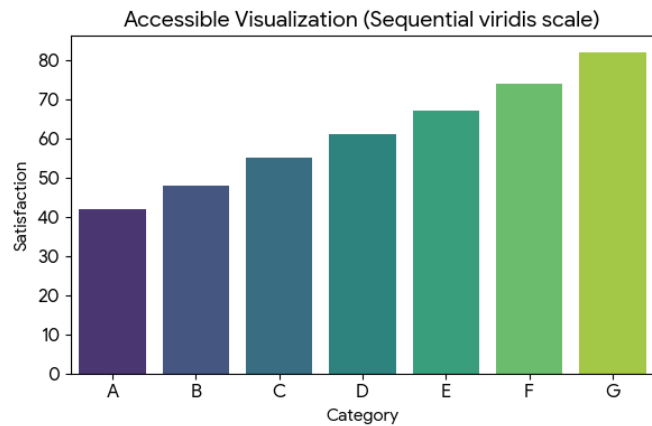
Misleading: Diverging or random color scheme for sequential data

```
ggplot(df5, aes(x = cat, y = sat, fill = cat)) +
  geom_bar(stat = "identity") +
  scale_fill_brewer(palette = "Set1") + # Non-monotonic
  labs(title = "Misleading Visualization (Categorical Colors)")
```

Redesign: Ordered sequential scale

```
ggplot(df5, aes(x = cat, y = sat, fill = sat)) +
  geom_bar(stat = "identity") +
  scale_fill_viridis_c(option = "viridis") +
  labs(title = "Redesigned Accessible Visualization (Sequential Scale)")
```





Critique: In the first plot, assigning contrasting categorical colors to naturally ordered categories confuses the viewer, as it suggests the categories are distinct rather than sequentially ranked by satisfaction. Moreover, some color combinations might be indistinguishable to colorblind users. The redesigned plot uses a perceptually uniform, colorblind-friendly sequential scale ('viridis'). This visually reinforces the magnitude of satisfaction scores and ensures accessibility.

Suggested R packages: ggplot2, dplyr, boot (or base R).

Assessment focus: Correct R code, appropriate uncertainty visualization, interpretation, and identification of misleading design choices.